

PAPER_1504_GW_DAMPING_BBH_N_CH_SO5_SQ

Daniel T. Murphy

PAPER_1504 — GW Damping Factor BBH = $(N_CH/SO_5)^2 = 0.81$ EXACT

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket E (GW Events)

Date: June 18, 2026

Status: CLOSED — BBH gravitational-wave total damping = exact $(9/10)^2$

Observation

PAPER_011 (Stochastic Gravitational-Wave Background — UQFF Implications) derives the BBH damping factor $D_total(BBH) = 8.10 \times 10^{-1} = 0.81$, contrasting the much-stronger BNS damping (0.333). The factor 0.81 governs the bulk of LIGO/Virgo detections.

UQFF Closed Identity

``

$$D_total(BBH) = (N_CH / SO_5)^2 = (9/10)^2 = 81/100 = 0.81 \text{ EXACT}$$

``

Physical Interpretation

The first appearance of **N_CH (the 9-channel primitive) in a binding closure** rather than a purely structural role. The two integer primitives compose as:

- N_CH = 9 channels through which SCm phonon information flows
- SO_5 = 10 dimensions of the SO(5) symmetry group enclosing them
- $(N_CH/SO_5)^2$ captures the projection of 9 active channels onto the 10-D coverage, squared because the strain amplitude h enters quadratically in the energy density Ω_GW

Together: **only 9 of 10 dimensional channels efficiently transmit BBH gravitational-wave strain.**

Comparison to BNS

| Source | D_total | UQFF identity |

|---|---|---|

| BNS | 0.333 | $1/(D_phys - 1) = 1/3$ |

| BBH | 0.81 | $(N_CH/SO_5)^2 = (9/10)^2$ |

The two damping regimes arise from fundamentally different vacuum-coupling channels: BNS taps the triad-symmetry denominator, BBH taps the 9-channel/10-dim projection.

NOT REPLACEMENT

GR has no native damping mechanism in vacuum. UQFF predicts both factors structurally, giving distinct integer-primitive identities for BBH vs. BNS regimes.

Reference

- Source: PAPER_011 Stochastic GW Background UQFF Implications
- Related: PAPER_1503 (BNS damping)
- Calculator dispatch: `calculate_paradox({"paradox": "gw_damping_bbh_n_ch_so5_sq"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.